Neal **Miller**

Engineering Director · AI-First Engineering · Product Delivery · Architectural Transformation & Modernization

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Summary

Engineering leader with a proven track record building and scaling multi-team engineering organizations — from leading ~8-person teams to a ~60-person organization at Imagine Learning alone — across complex, high-stakes product environments. That includes founding Imagine Learning's first cross-product platform engineering team, run deliberately lean and AI-leveraged. Currently Director of Software Engineering for Imagine Learning's market-leading courseware portfolio, EdgeEX and Edgenuity, across 9 teams, with a focus on platform migration, modernization, and long-term technical strategy. A practitioner and organizational champion of agentic, AI-first development, having led an early end-to-end adoption of this model in a commercial engineering organization. Ph.D., Columbia University.

Core Competencies

Leadership & Strategy	Multi-team management, Technical strategy, Hiring & talent development, Budget & vendor management, Platform migration, AI-first SDLC transformation
Technical Expertise	Service-oriented architecture (SOA), Event-driven systems, Distributed systems, Cloud-native modernization, Architectural governance
AI & Engineering Practice	Agentic development workflows, AI-augmented delivery, Engineering productivity, Claude Code / GitHub Copilot

Experience

Director of Software Engineering — Remote

Imagine Learning — October 2026 - Present

Promoted to Director of Software Engineering in October 2026, expanding scope from EdgeEX to all of **Courseware** engineering: **EdgeEX**, the next-generation high school platform, and **Edgenuity**, the established platform it is succeeding, together serving ~**2M** students. Lead 9 teams of ~60 engineers and contractors (~25 FTEs, ~35 contractors), reporting directly to the CTO: one manager over 3 teams and 6 teams led directly, including a distributed team based in the US and India.

Leadership & Delivery

- Own engineering for Courseware Prime, the program migrating Edgenuity users to EdgeEX (the central engineering objective for 2027) while solidifying the Edgenuity architecture EdgeEX depends on, with work underway to decommission its heaviest technical debt.
- Own the Courseware engineering budget: set the FTE/contractor mix, manage vendors, and adjust spend to corporate targets.
- Drive cross-team architectural alignment, dependency management, and delivery execution across both platforms.
- Champion adoption of AI-assisted development tools (Claude Code, GitHub Copilot) across the organization.

Senior Software Engineering Manager — Remote

Imagine Learning — June 2025 - October 2026

Promoted to Senior Software Engineering Manager in June 2025 to lead the **Literacy Suite** (3 teams, ~20 engineers, led through a mix of direct reports and a team-level manager). Moved in May 2026 to lead engineering for **EdgeEX**, Imagine Learning's next-generation courseware platform and the successor to its legacy Edgenuity platform, across 4 teams of ~32 engineers and contractors with ~14 direct reports.

EdgeEX (May 2026 - October 2026)

- Led EdgeEX through a successful 2026 back-to-school season, with production incidents and field-reported bugs each down ~30% year over year.
- Championed adoption of AI-assisted development tools (Claude Code, GitHub Copilot) across EdgeEX teams; comparable teams using them saw user-story delivery increase ~107% versus Q4 2025.
- Onboarded 7 engineers through Imagine Learning's new India office expansion.

Learning Platform (founded April 2026, handed off August 2026)

- Founded Imagine Learning's first cross-product platform engineering team to build a set of shared services supporting a new Core product and, more broadly, Imagine Learning's entire product suite.

- Built it as a deliberately lean, AI-leveraged team and the company's first adopter of an agentic, AI-first development model, set its delivery path, and handed it off to its current manager.

Literacy Suite (June 2025 - May 2026)

- Led the initiation of multi-year modernization work on legacy monoliths within the Literacy Suite, handing it off to continue under new leadership on transitioning to EdgeEX.
- Promoted a Literacy Suite engineer (early 2026).

Software Engineering Manager — Remote

Imagine Learning — June 2023 - June 2025

Led the Literacy Data & Insights team, delivering real-time educator-facing reports within the Literacy Suite, which serves **1M+** students.

- Delivered the Growth Report for BTS 2024 — the most technically complex report in the Literacy Suite — spanning district, school, group, and student views on a new real-time data architecture.
- Managed a team of up to ~8 software engineers and SDETs (mix of FTEs and contractors).
- Contributed to hiring decisions for both FTEs and contractors, and managed vendor relationships.

Software Engineering Manager — Remote

SmileDirectClub — June 2022 - June 2023

Promoted to Software Engineering Manager to lead the Core Services team, building foundational internal services supporting user-facing products across SmileDirectClub's engineering organization, including contributing infrastructure supporting **SmileOS** — the company's AI-powered treatment planning platform serving 1.5M+ customers.

- Managed a blended team of ~6 engineers and contractors.

Software Engineering Team Lead — Remote

SmileDirectClub — August 2020 - June 2022

- **Partner Network**
  - Transformed team delivery practices — eliminating dysfunctional sprint ceremonies and transitioning from multi-hour nightly releases to on-demand deployments with minimal overhead.
  - Led development of the Partner Portal supporting SDC's Partner Network initiative.
- **Prescriptions**
  - Led the transition of the Prescriptions domain from a monolithic Django application to an event-driven, service-oriented architecture.
  - Delivered the Treatment Plan Service and shifted all functionality with zero downtime and minimal disruption.

Senior Lead Software Developer — Houston, TX

FranchiCzar — March 2019 - August 2020

Led technical direction for a 6-person development team, driving architecture, delivery, and hands-on development across multiple products.

- Played a central role in hiring and onboarding of engineering team members.
- Oversaw development of FranchiCzar OS (franchise CRM), Fund.io (payments and subscriptions), and Rylos (LMS).

Earlier Engineering Roles — Houston, TX

Gene by Gene, Durwella, RigMinder — June 2014 - March 2019

Software Engineer Team Lead, Gene by Gene (November 2017 - March 2019); Senior Software Developer, Durwella (February 2016 - November 2017); Lead Software Engineer, RigMinder (June 2014 - February 2016).

Learning Analytics Engineer — Houston, TX

Reasoning Mind — July 2012 - June 2014

Developed algorithms for educational data mining within the Reasoning Mind adaptive learning platform, co-authoring peer-reviewed research on automated detection of student engagement and teacher behavior across a cohort of 70,000 students.

Education

Columbia University in the City of New York — New York, NY

Ph.D. Chemical Physics (with Distinction) — 2012

- Dissertation: *Self-organization in systems of anisotropic particles*

The University of Texas at Austin — Austin, TX

B.S. Chemistry, Honors (with Highest Honors) — 2007